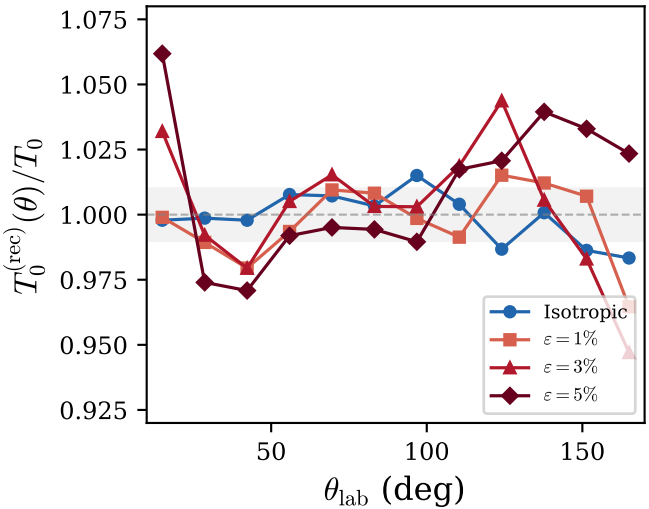
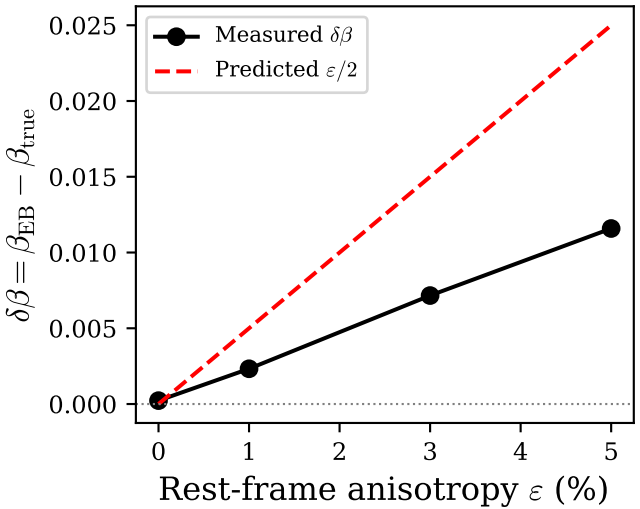
Adversarial robustness tests: HIGGINS parameters ($\gamma = 1.5$, $T_0 = 1\,\mathrm{keV}$, $N_{\mathrm{ens}} = 500$)

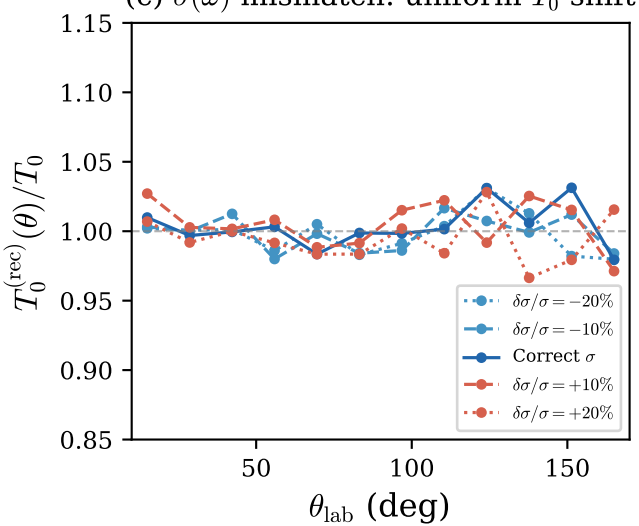
(a) Anisotropy: angular trend in $T_0^{(\mathrm{rec})}$



(b) Velocity bias from anisotropy



(c) $\sigma(\omega)$ mismatch: uniform T_0 shift



(d) Quantitative summary

ε	δβ meas	δβ pred	T_0 RMS
0%	+0.0002	+0.0000	0.92%
1%	+0.0023	+0.0050	1.42%
3%	+0.0072	+0.0150	2.43%
5%	+0.0116	+0.0250	2.63%

σ(ω) mismatch:

δσ/σ	δT_0/T_0	T_0 RMS	β shift
-20%	-0.18%	1.42%	+0.0005
-10%	-0.14%	1.20%	-0.0002
+0%	+0.33%	1.49%	+0.0001
+10%	+0.51%	1.63%	-0.0003
+20%	-0.56%	1.64%	-0.0000

Combined (ε=3%, δσ/σ=+20%):
δβ = +0.0065, T_0 RMS = 2.20%